

PCB Design Checklist

Contacts

Signal Integrity Team

Checklist to help PCB and system designers complete a PCB.

Layers

- > [At least one Ground Plane present.](#)
- > [Dedicated plane for VCCo.](#)
- > [Every signal layer is adjacent to or one layer away from a continuous reference plane.](#)

Bypassing

- > [High-frequency capacitors within 1 to 2 cm of each VCC pin.](#)
- > [Mid-frequency capacitors within 8 cm of VCC pins.](#)
- > [Low-frequency capacitors anywhere on PCB.](#)
- > [Bypass capacitor on each Vref pin.](#)

Simultaneously Switching Outputs

- > [SSO guidelines in datasheet have been checked.](#)

Signal Paths (PCB traces and terminations)

- > [Each trace has constant impedance.](#)
- > [Traces longer than \$T_r/6\$ have been simulated.](#)
- > [If ringing or overshoot is observed, add termination or change IO standard.](#)
- > [Extra attention has been paid to clock signals \(GCLK, CCLK, TCK etc\).](#)
- > [Long, closely-spaced parallel traces have been analyzed for crosstalk.](#)

Power Supply and Power Management

- > [Total FPGA power estimated with Power Estimator or XPower](#)
- > [Power supply satisfies POR monotonicity and ramp rate requirements](#)
- > [Power supply satisfies POR minimum current requirement](#)
- > [Die temperature predicted by \$T_j = T_a + P \cdot Q_{ja}\$ is less than max allowable](#)

Design for Debug

- > [JTAG header included on board \(connected to JTAG pins of device\)](#)

At Least One Ground Plane Present

Every board must have at least one continuous ground plane to:

- > Provide a low-impedance power system.
- > Make the connection of low-impedance vias to device GND pins very convenient.
- > Provides a path for return currents.

All of these are very important in for keeping ground noise at a minimum, both on the board and in the devices.

Dedicated Plane for VCCo

Including an optional VCCo plane (continuous, dedicated, or semi-dedicated) greatly simplifies board layout. It increases the accessibility of routing connections to power pins and bypass capacitors and provides a low-impedance path for return currents.

Every Signal Trace is Within One Signal Layer of a Continuous Reference Plane

Every trace in the stackup should either be adjacent to a reference (power or ground) plane, or only separated from the closest reference plane by one signal layer. This ensures that return currents always travel as near as possible to their corresponding trace. Adjacent signal layers should run in perpendicular directions so that vertical and horizontal layers alternate. This limits crosstalk between signal traces of adjacent layers. To maintain constant impedance from layer to layer, adjust the trace widths for each layer, as in [Figure 1](#) below. See page 3 of [XAPP231](#) for information on calculating the characteristic impedance of a PCB trace based on its dimensions, or use a field solver. It is important to maintain continuity of the reference plane. Signal traces should never cross discontinuities (large hole, slot, or break) in their associated reference plane.

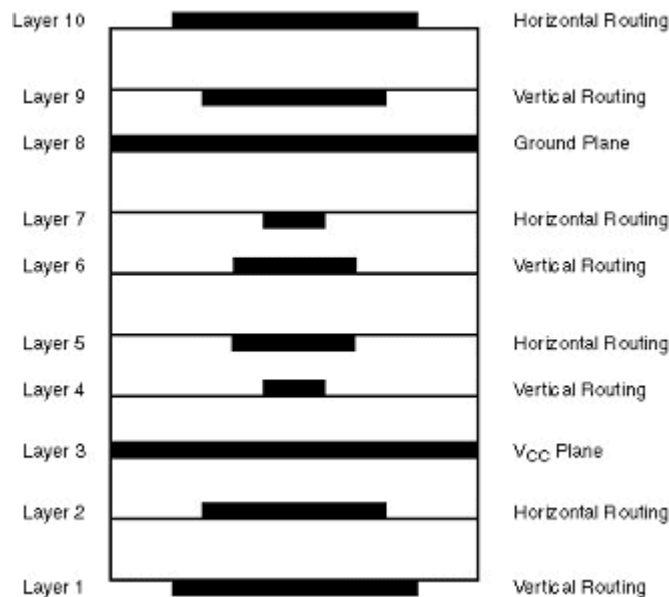


Fig. 1

High-frequency Capacitor Within 1 to 2 cm of Each VCC pin

High-frequency bypass capacitors are the smallest capacitors in the bypassing network. There should be at least one high-frequency capacitor on every VCC/GND pair, mounted within 1 to 2 centimeters of the VCC pin it is bypassing. The best location for these capacitors is underneath the PCB, directly beneath the FPGA.

Capacitor vias should **never** be shared. Each capacitor requires at least 2 vias connections: one ground, and one Vcc. Vias should descend directly to the power and ground planes (Do not use traces to connect bypass capacitors to the power pins they service).

The total capacitance of all high-frequency capacitors must be at least 25 times the equivalent switched capacitance ($C = P/(FV^2)$ for VCCint, $C = CLOAD \cdot N$ for VCCio). For greater noise immunity, a factor of 50 or 100 should be used instead of 25. When one capacitor per VCC/GND pin is used, this calculation usually results in a range of 0.1 μ F to 0.01 μ F. Smaller values of 0.0047 μ F and 0.0033 μ F should also be used.

All high-frequency capacitors should be low ESR ceramic chips. Always use the smallest package for a given capacitor value. See page 2 of [XAPP623](#) for more information on choosing capacitor size and characteristics. See capacitor vendors' websites for specific information on capacitor characteristics (<http://www.avxcorp.com/>, <http://www.tdk.com/>).

Mid-frequency Capacitors Within 8cm of VCC Pins

Mid-frequency bypass capacitors are low ESR, low inductance capacitors ranging from 4.7 μ F to 47 μ F. Tantalum capacitors are ideal; aluminum electrolytic capacitors may also be used. Use at least one for every 3000 slices (two for V4000, four for V10000, 7 for V20000E).

Low-frequency Capacitors Anywhere on PCB

Low-frequency bypass capacitors for board bypassing range from 47 μ F to 4700 μ F. For this function, any type of capacitor may be used anywhere on the board.

Bypass Capacitor on Each Vref Pin

Because of their high input impedance, Vref pins are susceptible to noise coupled in from surrounding signals. Every Vref pin needs a local bypass capacitor ranging from 0.01 μ F to 0.1 μ F. Noise from the power supply is not a concern, so do not use inductors or ferrite beads here.

SSO Guidelines have been Checked

See the datasheet for SSO guidelines. Multiply the number of effective VCC/GND pairs in your device (value in a chart, by device/package combination), by the SSO guideline number (value in a chart, by IO standard), to find out the total number of outputs that may safely be driven. Compute this on a bank by bank basis. Exceeding these guidelines can lead to serious ground bounce problems.

Each Trace has Constant Impedance

Every signal trace should maintain the same impedance no matter where it goes. Signal traces may be of any practical impedance value (40 ohms to 100 ohms is common). The same design may have signal traces with a variety of different impedance values. However, a single trace should not change impedance values over its length. For example, if a trace starts on one board layer and switches to another layer, the designer must ensure that the trace on the second layer has the same impedance as the first. If the layers are different distances from their respective reference planes, the widths of the signal traces should be adjusted

accordingly. In general, if the distance to the reference plane is increased, the trace width must also be increased in order to maintain the same impedance. See [Figure 1](#) above.

Traces Longer Than $T_r/6$ have been Simulated

The ratio of signal rise/fall time to trace length can determine whether or not transmission-line effects will occur. In general, long traces with fast rise/fall times exhibit transmission-line effects. If the time it takes a signal to propagate down the length of the trace is more than $1/6$ of the signal rise/fall time, transmission-line effects are likely, and the signal path must be simulated. This can be performed in an IBIS or SPICE simulator. For more information on transmission-line effects and simulation, see the text references at the end of this document.

If Ringing or Overshoot is Observed, Add Termination or Change IO Standard

A simulated transmission line that exhibits ringing or overshoot indicates an unacceptable amount of signal reflection. Signal reflection occurs when a signal wave encounters an impedance discontinuity. To repair the ringing or overshoot, you must eliminate the impedance discontinuity in one of three ways:

- Add resistive termination to the PCB (in series or parallel).
- Change the SelectIO standard to one with a lower current drive.
- Use XCITE DCI (in Virtex-II).

For further information on termination, see [our resources page](#).

Extra Attention has been Paid to Clock Signals (GCLK, CCLK, TCK etc)

Clock signals require special attention for two reasons. First, it is critical that their timing not be marginalized by noise - this can lead to false clocking of data. Second, clock signals often run at a higher frequency than data; they can be more troublesome as noise sources. Clock traces and their drivers should ALWAYS be carefully simulated prior to PCB fabrication.

Long, Closely-spaced Parallel Traces have been Analyzed for Crosstalk

Pay attention to any traces running in parallel for long distances. Simulate any suspicious traces using a PCB crosstalk simulation tool to determine if they will cause problems. If you confirm crosstalk as a problem, manage it by separating the traces or decreasing their distance from the associated reference plane (decrease dielectric thickness).

Total FPGA Power Estimated with Power Estimator or XPower

The [Power Estimator](#) or XPower should be used to determine the approximate power the FPGA will require. The Power Estimator requires design data generated by MAP (CLB utilization, Flops, IO standard, BlockRAM usage). XPower is part of the design flow. These tools provide a guideline for power supply requirements and are essential for thermal planning.

Power Supply Satisfies POR Monotonicity and Ramp Rate Requirements

The power supplies should rise from less than 0.1 Vdc to the minimum DC operating condition voltage level in less than 50 milliseconds, and not faster than 1 millisecond. The rise should not be inhibited by a current trip, or current foldback. Current limit behavior is acceptable based on the "Power-On Ramp Up Current Requirement" specification from the data sheet. The voltage rise vs. time should be roughly monotonic. Avoid dwelling at a voltage, or having a 'plateau', even if it is acceptable power supply behavior. If the voltage increases beyond the minimum operating voltage and then drops below it, you will experience incorrect power behavior. If the power supply voltage falls below the absolute minimum operating voltage when the device is off, it should not rise immediately back to the nominal operating voltage when turned on without first discharging to below 0.1 Vdc. You may need a resistor to bleed off the filter and bypass capacitor charges to meet this condition.

Power Supply Satisfies POR Minimum Current Requirement

Aside from meeting the dynamic power requirements determined by the [Power Estimator](#), the power supply must also be able to supply the minimum specified startup current specified in the datasheet.

Die Temperature Predicted by $T_J = T_A + P \cdot QJA$ is Less than Max Allowable

Determine the die temperature using the power figure derived from the Power Estimator, information about the device package, and the maximum ambient temperature in the operating environment. If it is higher than the maximum allowable temperature for the device temperature grade (C = Commercial: 0 °C - 80 °C, I = Industrial: -40 °C - 100 °C), the design must change (lower ambient temperature, add heatsink, change package, reduce clock frequency or reduce device utilization). Further information on thermal planning and management is located with [Advanced Packaging](#) and on page 1 of [XAPP623](#).

JTAG header included on board (connected to JTAG pins of device)

Every PCB should have easy access to FPGA JTAG pins. This enables debugging in the final system. For best results, route the TCK, TMS, TDI and TDO signals to a four-pin header on the PCB. This is critical with BG and FG packages which have limited access to device pins. You may also provide ground and VCC pins in the header for convenience (six pins).

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